VU AI HUB · LECTURE NOTES

History of AI: exam practice answers

Year 1, April to May 2026

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Exam XB_0113 2025 II

Section A: Lectures

1. **Prehistory of AI**

   The prehistory of AI shares common foundational events with the prehistory of computing, specifically in the quest to formalize and automate processes. While computing's prehistory focused on automating administration and mathematical calculations (such as Babbage's engines and the US Census Hollerith machines), the prehistory of AI focused on automating human thinking and language. Both histories converged around Boolean logic, which expressed the 'laws of thought' in mathematical True/False calculations, and Alan Turing's theoretical concept of a machine that could compute any logical deduction.

2. **Paris Conference**

   The January 1951 Paris conference on 'calculating machines and human thinking' showcased cybernetic innovations like William Grey Walter's electronic 'Tortoise', which simulated animal behavior. This suggests that the desire to emulate life and human intelligence was a primary intellectual driver for building early computers, demonstrating that scientists were entering the computing field specifically to explore artificial intelligence and behavioral simulation, not just to solve mathematical defense equations.

3. **Acronyms**

   In 1950s computing, acronyms served to give complex, room-sized machines a specific identity and functionality that outsiders could grasp. Acronyms like ENIAC, ZEBRA, and PETER often highlighted the machine's technical nature (ex. Binary, Electronic) or its national/institutional origins (such as Dutch PTT or Philips). By turning abstract engineering projects into recognizable names, acronyms acted as metaphors that made the new technology communicable to the public

4. **Grace Hopper**

   Grace Hopper made massive direct contributions to computing by pioneering automatic programming systems (compilers), such as the A-0 compiler and FLOW-MATIC, and playing a critical role in the development of the COBOL programming language. Indirectly, she contributed to AI by popularizing the metaphor that programming was 'educating a machine', drawing a direct conceptual link between machine coding and human cognitive learning.

5. **AI and Education**

   In the 1950s/60s, programmed instruction broke educational problems down into rigid, step-by-step processes modeled directly on the way computers process instructions (a behavioral approach of 'educating a machine'). In contrast, the 1970s/80s hype around LOGO utilized turtle graphics to teach children to actively write programs themselves, shifting the focus from simple imitation to deep understanding and problem-solving. Both hypes raised AI's visibility by connecting machine logic to child psychology and learning.

6. **History of AI**

The history of AI helps define the field by revealing how its boundaries constantly shift depending on the era's technology and goals. Early on, AI was defined by cybernetics and psychological modeling (simulating brains or animal behavior). Later, history redefined AI around expert systems and logical rules in the 1980s, and today it is defined by stochastic models and neural networks. These historical shifts show that AI is not a static technology but a fluid concept defined by whatever researchers are currently trying to make computers achieve.

7. **Computers and ordinary life**